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SOLVED NUMERICALS – JUNE/JULY 2024 (CSE STREAM)

1(C) An optical fiber has refractive index of core and cladding of 1.55 and 1.50, respectively. Calculate its numerical aperture and angle of acceptance if it is kept in air.

$$N.A = \sqrt{n_1^2 - n_2^2}$$
$$N.A = \sqrt{(1.55)^2 - (1.50)^2}$$
$$N.A = 0.39$$
$$\theta_0 = \sin^{-1}(N.A)$$
$$\theta_0 = \sin^{-1}(0.39) = 23^\circ$$

2(C) In a diffraction grating experiment the Laser light undergoes first order diffraction at an angle of 19.3°. Find the wavelength of Laser light. Given the grating constant d = 1.98×10^{-6}m.

$$\lambda = \frac{d \sin\theta}{n}$$
$$\lambda = \frac{1.98 \times 10^{-6} \times \sin(19.3^\circ)}{1}$$
$$\lambda = 654 \times 10^{-9} m = 654 nm.$$

3(C) An electron is bound in an infinite potential well of width 0.18nm. Find its energy values in the first two allowed energy states.

$$E = \frac{n^2 h^2}{8ma^2}$$
$$E_1 = E_0 = \frac{1^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (0.18 \times 10^{-9})^2} = 1.86 \times 10^{-18} J$$
$$E_2 = 4E_0 = 4 \times 1.86 \times 10^{-18} = 7.45 \times 10^{-18} J$$

4(C) Calculate the kinetic energy of a neutron in eV. Given: de –Broglie wave length is 1 Å and mass of neutron, $m_n = 1.674 \times 10^{-27}$Kg.

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$\lambda = \frac{h}{\sqrt{2mE}}$ Squaring & rearrange $E = \frac{h^2}{2m\lambda^2}$

$E = \frac{(6.63\times10^{-34})^2}{2\times1.674\times10^{-27}\times(1\times10^{-10})^2} = 1.3\times10^{-20}$ J

$E = \frac{1.3\times10^{-20}}{1.6\times10^{-19}} = 0.082$ eV

6(C) A linear operator 'X' operates such that $X|0\rangle = |0\rangle$ and $X|1\rangle = i|1\rangle$. Find the matrix representation of 'X'.

$X|0\rangle = |0\rangle$

$\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

i.e $\begin{pmatrix} x_{11} \\ x_{21} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow x_{11} = 1,\ x_{21} = 0$

$X|1\rangle = i|1\rangle$

$\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = i\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\begin{pmatrix} x_{12} \\ x_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} \Rightarrow x_{12} = 0,\ x_{22} = i$

$\therefore X = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

7(C) A lead wire has a critical field of 6.5×10^3 A/m at 0 Kelvin. The critical temperature is 7.18K. At what temperature the critical field will drop to 4.5×10^3 A/m.

$H_c = H_0\left[1 - \frac{T^2}{T_c^2}\right]$

$4.5\times10^3 = 6.5\times10^3\left[1 - \frac{T^2}{7.18^2}\right]$

Simplify

$T = 3.98$K

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8(C) Calculate the probability of occupation of an energy level 0.02eV above level at temperature 27°C.

$$f(E) = \frac{1}{1+e^{E-E_F/KT}}, \qquad T = 273+27 = 300k$$

$$= \frac{1}{1+e^{\frac{0.02\times1.6\times10^{-19}}{1.38\times10^{-23}\times300}}} = 0.316$$

9(C) In the case of animating a jump, the jump height is 2.5m and jump magnification is 5. Calculate the push height and push acceleration. Given gravitational acceleration is 10m/s.

$$JM = \frac{\text{Jump height}}{\text{push height}}$$

$$\text{Push height} = \frac{\text{Jump height}}{JM} = \frac{2.5}{5} = 0.5m$$

$$JM = \frac{\text{Push acceleration}}{\text{gravitational acceleration}}$$

$$\text{Push acceleration} = 5\times10 = 50 m/s^2$$

10(C) On a particular place, volcanic eruption occurs once in every 100years on an average. Calculate the probability of volcanic eruption in a 100 years interval for K = 0, 1 and 2, assuming the Poisson's model appropriate.

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$P(X=k) = \frac{1^k e^{-1}}{k!}$$

$$P(X=0) = \frac{1^0 e^{-1}}{0!} = 0.368$$

$$P(X=1) = \frac{1^1 e^{-1}}{1!} = 0.368$$

$$P(X=2) = \frac{1^2 e^{-1}}{2!} = 0.184$$

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